

Role Requirements as Sources of Organizational Stress

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Relationships between major role requirements and experienced role stress were examined on the basis of data drawn from 202 research and development professionals. Measures of role stress included various types of role conflict and ambiguity. Role requirements included integration and boundary-spanning activities, personnel supervision, and scientific research. Role conflict appeared to be more sensitive than role ambiguity to differences in research and development role requirements; and integration and boundary-spanning activities were the best predictors of experienced role conflict, especially of the inter-sender variety.

There is growing evidence which suggests the potential of role theory as a framework for understanding the interactions and outcomes of individuals in complex organizations; but empirical studies have been few in number and have seldom focused on differences in role perceptions as a function of major role requirements. Previous research has demonstrated that both role conflict and role ambiguity are adversely related to a host of work-related outcomes (e.g., Gross, Mason, & McEachern, 1958; House & Rizzo, 1972; Ivancevich & Donnelly, 1974; Kahn, Wolfe, Quinn, Snoek, & Rosenthal, 1964; Lyons, 1971; Organ & Greene, 1974); but only one of these studies included an attempt to distinguish role perceptions on the basis of differences in role requirements (Kahn et al., 1964).

Across these studies, degree of role conflict or role ambiguity has been shown to be related directly to job-induced tension and anxiety and propensity to leave the organization and inversely related to job satisfaction and attitudes (e.g., trust, respect, and liking) toward role senders. Furthermore, various

types of role conflict, including intersender conflict and person-role conflict, have been identified (Kahn et al., 1964) and operational definitions and scales for their measurement have been reported (House & Rizzo, 1972); but no attempt has been made to relate these conflict types to specific organizational roles. Thus it is likely that persons occupying different roles may experience equivalent degrees of role conflict, but the sources and specific types of conflict they experience may be quite different. This same shortcoming exists for role ambiguity studies, which have dealt with the general measure without focusing on the specific types of ambiguity (e.g., ambiguity regarding role expectations or ambiguity regarding performance evaluation). More work needs to be done to explain the occurrence of these types of experienced role ambiguity on the basis of different role requirements imposed upon the individual.

In summary, previous research has not looked beyond the *general* role stress variables to discover the specific nature and sources of role conflict and ambiguity in complex organizations. Furthermore, different role requirements existing within the same organizational context have not been related to experienced role stress. Until some effort is made to close these gaps, the diagnosis of organizational stress is likely to be of little value in understanding employee strain and in designing mechanisms (e.g., selection and placement, job/role design, supervisory training, etc.) to manage its dysfunctional consequences for individuals and organizations.

This study was conducted as part of a dissertation, under the direction of Professor J. Stacy Adams, in partial fulfillment of the requirements for the doctoral degree at the University of North Carolina, Chapel Hill. Portions of this article were presented at the 35th Annual Meeting of the Academy of Management, New Orleans, August, 1975.

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Role Requirements as Predictors of Stress

There is evidence to suggest that objective role requirements may be useful in predicting the degree of role *conflict* experienced by a role incumbent. One major role requirement which is believed to contribute to experienced role conflict, and perhaps role ambiguity, is the extent of boundary-spanning activities. Persons engaged in these activities must maintain a delicate system of linkages across differentiated systems or subsystems and this linking function is viewed as a major source of strain and conflict in complex organizations (Corwin, 1969; Thompson, 1960).

Under these conditions, *integrator* positions (Lawrence & Lorsch, 1969) may be established in highly differentiated organizations to supplement more conventional control and coordination mechanisms, and *boundary role persons* (Adams, in press) may be found coordinating interdisciplinary or cross-functional activities both within and across organizational boundaries. These coordinative responsibilities across boundaries are potential sources of role conflict because they place the role incumbent in a position where he/she is the target of influence attempts originating in different organizations or subsystems. Furthermore, the conflict potential of these boundary-spanning role requirements should increase with the level of differentiation between these units. This condition is viewed as leading to potentially higher levels of role conflict (especially intersender conflict) for the boundary role person than for individuals occupying internal organizational roles.

Kahn et al. (1964) also regard roles which are high in *boundary relevance* as having high potential for chronic role conflict. They reported that boundary relevance, as measured by the self-reported frequency of contact with persons outside the organization, was directly related to the level of experienced role conflict. They also reported that frequency of interdepartmental contacts was only slightly less related to experienced conflict than the frequency of interorganizational contacts. They did not distinguish the level of role conflict experienced by the boundary role person from persons occupying other orga-

nizational roles or pursuing other major role requirements. This appears to be needed if we are to know how much *more* conflict is associated with boundary versus internal organizational roles.

Relatively high levels of role conflict have also been thought to be associated with supervisory role requirements (Charters, 1952; Roethlisberger, 1945), and evidence concerning the inherent conflict in supervisor-subordinate relations in research and development organizations has been reported (Kaplan, 1959; LaPorte, 1966). Thus, role conflict appears to be associated with organizational linking roles, whether they involve the integration of activities across intraorganizational and interorganizational boundaries or up and down the hierarchy. Persons occupying interunit and interorganizational boundary-spanning roles between differentiated systems may be expected to experience greater degrees of role conflict than persons linking different levels of a hierarchy within the same organizational context. On the other hand, persons occupying internal, buffered roles, especially nonsupervisory scientists and engineers, would not be as exposed to conflicting pressures as persons in linking roles.

Previous research has generally not focused on role requirement differences in experienced role ambiguity. Kahn et al. (1964), for example, did not report the relationship between their measure of frequency of extra-organizational contacts and role ambiguity. One generalization, however, has been that managerial roles are more ambiguous than nonmanagerial roles. A test of this proposition was not possible using the present sample since the nonmanagerial roles were composed of basic and applied scientists and engineers whose roles are inherently ambiguous. Therefore, a parallel, but exploratory, study of relationships between professional-level research and development role requirements and role ambiguity was conducted.

Based on this evidence, the following hypotheses were tested in the present investigation:

Hypothesis 1. The extent of integration and boundary-spanning role requirements is directly related to the level of role conflict

(particularly intersender conflict) experienced by focal persons.

Hypothesis 2. The extent of supervisory role requirements is directly related to the level of role conflict experienced by focal persons though the strength of the relationship is not as great as that obtained for integration and boundary-spanning activities.

Hypothesis 3. The extent of both internal and external organizational linking role requirements (i.e., integration and boundary-spanning activities and supervisory activities) is more strongly related to experienced role conflict than the extent of highly buffered, internal organizational role requirements which are nonsupervisory in nature (i.e., scientific research activities).

METHOD

Research Site and Sample

In order to test these hypotheses, it was necessary to obtain host organizations which could provide access to a number of persons occupying roles which were high in boundary relevance and which were located in the same organizational context with persons in more conventional, supervisory and nonsupervisory, internal roles. Nine research and development directorates located in the Southeast met these criteria. They were semiautonomous directorates of a large laboratory, having parallel role structures but being highly differentiated from one another in terms of professional specialization and relative emphasis on basic versus applied research.

Of the 210 professionals asked to volunteer for participation in this study, 96% completed usable questionnaires. These participants represented five major organizational roles: integrator ($n=28$), division manager ($n=43$), technical group leader ($n=38$), nonsupervisory basic scientists and engineers ($n=30$), and nonsupervisory applied scientists and engineers ($n=63$). The job classification of "integrator" in this study includes both boundary role persons (Adams, in press) and intraorganizational integrators (Lawrence & Lorsch, 1969). They are personnel employed in roles which require them to coordinate the efforts of others located in different directorates or in organizations outside the boundaries of the laboratory. They hold the following job titles: advanced research projects manager, advanced systems concepts manager, and lead system engineer. They have in common the role requirement of coordinating interdisciplinary and cross-functional projects across intraorganizational and interorganizational boundaries. Division managers are second-level (from the top) managers at the directorate level. Group leaders occupy the third managerial level within the directorates and are assigned technical, but not administrative support, supervisory responsibilities.

The distinction between basic and applied engineers is based on their organizational membership in either a *basic* or *applied* research directorate.

Procedure

Participants completed a questionnaire administered by the researcher in approximately 1½ hours in conference rooms located near their places of work.

Instruments

Role stress variables. Role conflict was defined as the degree of incongruity or incompatibility of expectations a focal person experiences in the performance of an assigned role. The Rizzo, House, and Lirtzman (1970) measure of *general* role conflict was used because of its demonstrated factorial independence from role ambiguity. *Specific* types of conflict were identified as subscales of this general measure. Intersender conflict occurs when the focal person perceives that demands from one role sender oppose demands from one or more other senders. Person-role conflict is the perceived incongruence between the role requirements placed on a focal person and his/her orientations, interests, and values. Intrasender conflict is perceived incongruence between various demands placed on the focal person by a single role sender, and role overload exists when the focal person cannot accomplish all of the role demands received.

Role ambiguity was defined as the lack of clarity of role expectations and predictability of role performance outcomes. This general measure was also developed by Rizzo et al. (1970). In addition, two *specific* measures of role ambiguity were developed. The focal person was asked to list (up to 10) important role senders which constituted his/her role set (Merton, 1957). Ambiguity regarding expectations was calculated by averaging the focal person's responses for each role sender to the following question: "As far as you know, does he usually let you know when he expects or wants something from you, or does he often keep these things to himself?" Ambiguity regarding role performance evaluations was based on the role-set average obtained for the following question: "Do you usually feel that you know how satisfied he is with what you do?" Both questions were drawn from a previous study by Kahn et al. (1964).

The number of items, means, standard deviations, and Spearman-Brown reliabilities for these role perception measures are presented in order in parentheses which follow: general role conflict (10, 2.79, .76, .84), intersender conflict (4, 2.73, .94, .81), person-role conflict (3, 2.64, .91, .84), intrasender conflict (2, 2.98, 1.19, .91) role overload (1, 3.03, 1.27), general role ambiguity (6, 2.68, .83, .92), ambiguity regarding role expectations (1, 2.02, .71), and ambiguity regarding role performance evaluations (1, 2.44, .67). All items were scored using a 5-point response mode.

Major role requirements. Rather than rely on a priori assumptions of differences in role requirements

based on job titles, an attempt was made to determine *actual* differences in role requirements. Each focal person was asked to rate the extent to which each of 50 job activities was a part of his/her role. Some of the supervisory activities were adapted from a longer list reported by Stogdill and Shartle (Note 1), while others, including boundary-spanning and scientific research activities, were added by the present researcher, who had formerly occupied the position of *both* integrator and special assistant to the director in these organizations.¹ These items were selected with the objective of covering the spectrum of activities which characterize research and development roles and reflected the researcher's prior experience in this type of organization.

It was anticipated that the following types of role requirements would appear in the factor solution: a supervisory component, a boundary-spanning or integration component, and a scientific research component. Each activity was rated on a 5-point scale ranging from "definitely not part of my job" to "a most significant part of my job." These ratings were factor analyzed using a statistical package for the social sciences (Nie, Hull, Jenkins, Steinbrenner, & Bent, 1970) principal component-method with varimax rotation to reveal the major underlying role requirements and to provide empirical justification for the item composition of those role requirement measures.

RESULTS

Development and Validation of Role Requirement Measures

The underlying role requirements were representative of the types of professional role requirements which characterize research and development organizations and were unambiguously anchored on the basis of organizational role. The first three factors (accounting for 46% of the common variance) were personal supervision activities (27.9%), integration and boundary-spanning activities (10.0%), and scientific research activities (8.5%). The criterion for inclusion of an item in a factor was that the item have a loading in excess of .40 on that factor and below .40 on all other factors in the solution. The factor matrix and the items included in each of the role requirement factors, along with their loadings, are presented in Table 1.

In addition to their representativeness, these role requirement factors were unambiguously anchored on the basis of organizational role. Integration and boundary-spanning activities were rated highest by integrators and lowest by buffered, basic scientists; and the extent of these activities increased systematically

through the roles from the highly buffered internal role of basic scientist, applied scientist and engineer, group leader, division manager, to integrator. Personnel supervision activities were rated highest by division managers, followed by group leaders; and scientific research activities were rated highest by basic scientists (see Table 2). Finally, the overall differences between roles on these requirements, as shown in Table 3, were significant (personnel supervision, $F = 107.06$, $p < .001$; integration and boundary spanning, $F = 21.28$, $p < .001$; and scientific research, $F = 11.26$, $p < .001$; and the estimates ($\hat{\omega}^2$) of the strength of statistical relations between roles and role requirements are strong enough to support the practical significance of these differences. The proportion of variance in role requirements accounted for by organizational role was as follows: personnel supervision (68%), integration and boundary spanning (28%), and scientific research (17%).

Role Requirements and Role Conflict

A correlational analysis of relationships between the role stress and role requirement variables provides support for the hypotheses (see Table 4). Both general role conflict and intersender conflict were directly related to the role requirement measure of integration and boundary-spanning activities. These conflict variables were also directly related to the supervision measure, but they were not correlated with the scientific research measure.

Differences in the relative strengths of correlations between role requirements and experienced conflict were also examined, using Hotelling's t test of differences in correlated correlation coefficients (Guilford & Fruchter, 1973; Hotelling, 1940). This test was used because the role requirement predictors of conflict were not independent (i.e., they were obtained from the same source). This test of differences takes into account the inter-correlations among predictors.

The differences in the role requirement correlations with general role conflict and

¹ A complete list of the 50 job activity items, referred to as ROLEREQS (R&D), can be obtained from the author.

TABLE 1
MAJOR ROLE REQUIREMENTS: RESULTS OF FACTOR ANALYSIS OF FIFTY JOB ACTIVITY ITEMS

Role requirement factors and items	Factor loadings			% vari- ance	Cumulative % vari- ance
	I	II	III		
Factor I: Personnel supervision activities					
1. Nominate personnel in the organization for promotion.	.89	.08	-.06		
2. Make decisions concerning the hiring and/or termination of employees.	.86	.09	-.05		
3. Counsel subordinates about their professional development.	.84	.12	.02		
4. Conduct regular staff meetings.	.84	.12	-.08		
5. Present awards to deserving employees.	.82	.15	.01		
6. Counsel subordinates about their personal problems.	.82	.12	.11		
7. Prepare personnel reports concerning subordinates.	.78	.23	.14		
8. Make assignments of jobs to subordinates.	.76	.15	.10		
9. Train new employees in performance of work.	.72	.16	.05		
10. Submit regular reports concerning accomplishments of groups of employees.	.67	.19	-.07		
11. Instruct subordinates in proper procedures.	.61	.04	.14		
12. Forecast the volume of work to be done in the near future.	.50	.34	-.17	27.9	27.9
Factor II: Integration and boundary-spanning activities					
1. Represent your directorate/laboratory to outsiders.	.25	.74	.01		
2. Review plans with agencies outside the laboratory.	.27	.72	-.03		
3. Travel at least 30 days per year as a representative of your organization	.09	.69	-.11		
4. Inspect or review projects under contract.	.17	.68	.19		
5. Integrate or coordinate interdependent activities of others.	.33	.58	-.06		
6. Prepare reports for governmental agencies outside the laboratory.	.27	.57	.34		
7. Request performance obligations from persons outside your immediate unit.	.18	.55	-.01		
8. Prepare and give several briefings per month.	.23	.51	.17		
9. Write or dictate at least 10 letters per week.	.31	.47	-.02		
10. Prepare contracts for approval by a superior.	.21	.46	-.02		
11. Act as liaison with other units of the organization.	.24	.41	-.18	10.0	37.9
Factor III: Scientific research activities					
1. Make contributions to a body of scientific knowledge.	.00	-.13	.84		
2. Write scientific/technical papers.	-.01	-.04	.74		
3. Personally conduct research.	-.09	-.23	.72		
4. Review manuscripts (technical).	.14	.31	.63		
5. Apply for patents and/or copyrights for work you have done.	.23	.00	.48	8.5	46.4

intersender conflict provided additional support for the hypotheses. The correlations between boundary-spanning activities and both role conflict and intersender conflict were significantly greater than those which were found for personnel supervision activities ($p < .05$ and $p < .01$, respectively). Thus, support was obtained for Hypothesis 2 which stated that role conflict, and particularly intersender conflict, is related more strongly to the extent of boundary-spanning activities than to super-

visory activities. Furthermore, the differences in these same conflict correlations with boundary-spanning and scientific research activities were significant and in the predicted direction ($p < .01$), favoring boundary-spanning activities for both general and intersender role conflict). Finally, the general and intersender conflict correlations with supervisory and scientific research activities were in the predicted direction and approached significance ($p < .10$ in both cases). Thus,

role requirements appear to do a good job of distinguishing the types and levels of role conflict experienced by professional-level personnel in research and development organizations. The best predictor of role conflict, and its intersender variety, was the integration and boundary-spanning role requirement.

Role requirement differences for person-role conflict were not observed between boundary-spanning and supervisory activities, but were significant between these two role requirements and scientific research activities. Finally, role overload and intrasender conflict did not appear to be distinguished on the basis of these professional-level research and development role requirements.

Role Requirements and Role Ambiguity (Exploratory Investigation)

In general, the measures of role ambiguity were not related to scientific research activities but were significantly and inversely related to both boundary-spanning and supervisory role requirements. Significant differences in the ambiguity-role requirement correlations were observed only for the ambiguity regarding role performance evaluations measure. In this case, the importance of both boundary-spanning and supervisory activities

TABLE 2
ROLE REQUIREMENT MEANS FOR
ORGANIZATIONAL ROLES

Organizational role	Role requirement factor means		
	Integration and boundary-spanning activities	Personnel supervision	Scientific research
Integrators	3.45 (1) ^a	1.64 (3)	2.38 (5)
Division managers	3.37 (2)	3.73 (1)	3.28 (3)
Group leaders	2.95 (3)	2.45 (2)	3.67 (2)
Applied scientists and engineers	2.38 (4)	1.58 (4)	3.17 (4)
Basic scientists and engineers	2.15 (5)	1.53 (5)	3.86 (1)
Total sample	2.81	2.21	3.29

^a Rankings of role means on extent of role requirement are shown in parentheses.

TABLE 3
ANALYSIS OF VARIANCE OF MAJOR ROLE REQUIREMENTS

Source	SS	MS	df	F ^a	Δ ^{2b}
Integration and boundary-spanning activities	49.71	12.43	4, 195	21.28*	28%
Personnel supervision	149.27	37.32	4, 195	107.06*	68%
Scientific research	38.76	9.69	4, 195	11.26*	17%

^a Comparisons of overall differences in role means on importance or extent of role requirement.

^b Δ² is the estimate of the strength of the statistical relation between organization role and extent of role requirement in a one-way analysis of variance (Hays, 1963, pp. 381-384).

* $p < .001$.

was inversely related to ambiguity regarding evaluations, and the correlations for both of these activities were significantly stronger than the correlation for research activities ($p < .05$ for both differences). Thus, the importance of certain role requirements to research and development professionals appears to be associated with less ambiguity concerning performance evaluation. Otherwise, this evidence suggests that role ambiguity is not very sensitive to differences in the role requirements of research and development professionals.

DISCUSSION

The focus of this investigation has been on the identification of predictors of stress in research and development organizations. Boundary relevance of the role occupied by a focal person was found to be the best predictor of experienced role conflict, especially intersender role conflict. Boundary relevance was inferred from the extent to which integration and boundary-spanning activity was a major role requirement placed on an organizational member. The extent of supervisory role requirements was also directly related to the degree of general and intersender role conflict, though the strength of relationship was not as great as that found between boundary relevance and these conflict measures. Finally, scientific research activities were not related to experienced role conflict. Thus, a pattern in the strength of relationships between major role requirements and experienced role conflict emerged, with highly buffered, scientific research activities bearing no relationship to experienced conflict and

TABLE 4
CORRELATIONS BETWEEN IMPORTANCE OF ROLE REQUIREMENTS AND VARIOUS TYPES OF ROLE
CONFLICT AND AMBIGUITY EXPERIENCED BY ROLE INCUMBENTS

Major Role Requirement Factors	Degree of role conflict					Degree of role ambiguity		
	Role conflict	Person-role conflict	Inter-sender conflict	Intra-sender conflict	Role overload	Role ambiguity	Ambiguity re expectations	Ambiguity re evaluations
Personnel supervision	.25***	.19**	.16*	.20**	.19**	-.19**	-.07	-.22***
Integration and boundary-spanning activities	.36***	.22***	.36***	.16*	.25***	-.14	-.15*	-.23***
Scientific research	.09	.05	.03	.10	.25***	-.10	-.08	-.02

Note. $N = 202$. Differences in correlations were evaluated using Hotelling's (1940) t test of differences in correlated correlation coefficients. A one-tailed test was employed for role conflict correlations, in accordance with the directional hypotheses, and a two-tailed test was used for the role ambiguity correlations which were part of the exploratory analysis. Level of significance was indicated by the following: $[] = p < .10$; $\{ \} = p < .05$; $\{ \} = p < .01$.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

organizational and interorganizational linking role requirements being significantly related to experienced role conflict. Furthermore, specific types of role conflict were shown to be differentially related to the role requirement predictors. This finding indicates that preoccupation with the general role conflict measure may obscure the real nature of conflict experienced, its cause, and its remedies. In summary, these results suggest that the degree and type of role conflict a person experiences on the job are contingent upon the major requirements of the role occupied.

Role ambiguity, on the other hand, does not appear to be as sensitive to differences in these particular role requirements. This finding, however, may be an artifact of the particular setting in which this research was conducted and caution should be used in attempting to generalize this finding to more conventional work organizations. While performance in managerial and boundary-spanning roles has generally been regarded as more ambiguous and, consequently, as more difficult to evaluate than the performance of nonsupervisory subordinates, this may not be the case in research and development (and other professional) organizations where the work of the scientist and nonsupervisory professional may be the most unstructured and ambiguous, and criteria of effectiveness the most difficult to establish.

Suggestions for Future Research

An assumption made in the present study, and in previous studies on role stress, is that particular role requirements are causally related to role stress variables, and an important step for future research on role stress would be to test this assumption. One way to approach this task would be to track persons as they move from one organizational role to another, monitoring the levels and types of role stress they experience over time. The same approach could be employed in a laboratory setting with greater ease and control, but perhaps less external validity.

There is also a need to look at relationships between multiple role requirements and role stress. For example, a particular role requirement may be unrelated to stress, but when added to other role requirements placed on a person may significantly increase the level of stress experienced.

The relationship between boundary relevance and role ambiguity should be tested in a more conventional work organization. The results with regard to this relationship in the present study may have been caused by the relatively high degree of ambiguity which permeates all professional research and development roles, regardless of organizational level.

Finally, it is likely that the concept of boundary relevance may be multivariate in

nature, consisting of different types of boundary-spanning activities. These activities may include representational, transactional, and integrative role requirements, as well as buffering, mediative, technological gatekeeping, and information transfer and filtration activities. While the purpose of the present investigation was to distinguish boundary spanning from other organizational role requirement predictors of stress, future studies may be directed at isolating these components of boundary relevance and determining their relationships with various types of role stress, dimensions of organizational environment and structure, and managerial perceptions and strategic choices.

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(Received June 19, 1975)